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# 👤 DEV 4 — PLAN (UPDATED: Frontend Dashboard + Map Intelligence | SynapseSignal)

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## 🔷 ROLE

Build the **Control & Visualization Layer** of SynapseSignal:

- Real-time dashboard
- Map-based traffic visualization
- Emergency vehicle tracking
- Green corridor visualization
- System control interface

Dev 4 ensures:

- Clear visualization of system state
- Real-time updates
- Interactive control capability
- Seamless integration with Dev 2, Dev 3, Dev 5

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## 🔷 PHASED EXECUTION PLAN

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### 🔹 PHASE 1 — FRONTEND SETUP

**Goal:** Initialize frontend environment

**Tasks:**

- Setup React project
- Install:
  - Leaflet (map)
  - Axios (API calls)
  - Tailwind CSS

- Create base folder structure

**Deliverable:**
✔ Working frontend skeleton

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## 🔹 PHASE 2 — MAP INTEGRATION (CORE)

**Goal:** Render real-world map

**Tasks:**

- Integrate Leaflet
- Load OpenStreetMap tiles
- Set:
  - default city view
- Enable:
  - zoom
  - pan

**Deliverable:**
✔ Interactive Map UI

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## 🔹 PHASE 3 — INTERSECTION VISUALIZATION

**Goal:** Display intersections on map

**Tasks:**

- Plot:
  - intersection nodes (lat/lon from Dev 2)
- Add:
  - markers
- Enable:
  - click interaction

**Deliverable:**
✔ Intersection Markers

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## 🔹 PHASE 4 — SIGNAL STATE VISUALIZATION

**Goal:** Show signal status

**Tasks:**

- Display:
  - RED / GREEN / YELLOW per intersection
- Update in real-time
- Color-code markers

**Deliverable:**
✔ Live Signal Visualization

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## 🔹 PHASE 5 — LANE & SECTOR OVERLAY

**Goal:** Show detailed intersection data

**Tasks:**
On click:

- Display:
  - lane density
  - sector scores
  - flow metrics

**Deliverable:**
✔ Intersection Detail Panel

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## 🔹 PHASE 6 — EMERGENCY VEHICLE TRACKING

**Goal:** Track emergency vehicles on map

**Tasks:**

- Render moving marker
- Update:
  - position (lat/lon)
  - velocity
- Highlight:
  - vehicle type

**Deliverable:**
✔ Real-Time Emergency Tracking

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## 🔹 PHASE 7 — ROUTE VISUALIZATION

**Goal:** Show selected route

**Tasks:**

- Draw polyline:
  - route path
- Highlight:
  - start → destination

**Deliverable:**
✔ Route Display

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## 🔹 PHASE 8 — GREEN CORRIDOR VISUALIZATION (CRITICAL)

**Goal:** Show active corridor

**Tasks:**

- Highlight:
  - corridor intersections
- Animate:
  - signal progression
- Color:
  - green path

**Deliverable:**
✔ Corridor Visualization

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## 🔹 PHASE 9 — ROUTE COMPARISON UI (ADVANCED)

**Goal:** Show route suitability

**Tasks:**

- Display:
  - current route
  - alternative routes
- Show:
  - ETA

    - congestion
- Highlight best route

**Deliverable:**
✔ Route Analysis Panel

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## 🔹 PHASE 10 — REAL-TIME DATA INTEGRATION

**Goal:** Connect frontend to backend

**Tasks:**

- Fetch:
    - traffic state (Dev 2)
    - decisions (Dev 3)
- Use:
    - polling or WebSockets

**Deliverable:**
✔ Live Data Sync

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## 🔹 PHASE 11 — CONTROL PANEL (OPTIONAL BUT POWERFUL)

**Goal:** Allow manual interaction

**Tasks:**

- Enable:
    - manual override
    - emergency simulation
- Send commands to backend

**Deliverable:**
✔ Interactive Control UI

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## 🔹 PHASE 12 — DASHBOARD ANALYTICS

**Goal:** Show system insights

**Tasks:**
Display:

- congestion levels

- traffic trends
- system status

**Deliverable:**
✔ Analytics Dashboard

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### 🔹 PHASE 13 — UI OPTIMIZATION

**Goal:** Improve performance & UX

**Tasks:**

- Optimize rendering
- Reduce re-renders
- Improve responsiveness

**Deliverable:**
✔ Smooth UI Experience

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### 🔹 PHASE 14 — ERROR HANDLING & FALLBACK UI

**Goal:** Handle failures gracefully

**Tasks:**

- Show:
  - loading states
  - error messages
- Handle:
  - API failure

**Deliverable:**
✔ Robust UI

---

## 🔷 DEV 4 SUCCESS CRITERIA

- Map renders correctly with real data
- Intersections displayed accurately
- Signal states update in real-time
- Emergency vehicles tracked live
- Green corridor clearly visualized

- Route comparison works correctly
- UI responds smoothly
- Backend integration stable
- Control panel works (if implemented)
- System is intuitive and interpretable

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## 🧠 FINAL INSIGHT

> Dev 4 is the **eyes of SynapseSignal for humans**

It transforms:

- complex system state
  into
- clear, interactive visual intelligence

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✔ Ready for:

- **Master Prompts (next)**
- **Guided Learning (after that)**

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# 👤 DEV 4 — MASTER PROMPTS (Frontend Dashboard + Map Intelligence | SynapseSignal)

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## 1️⃣ MASTER PROMPT — REACT PROJECT SETUP

You are a frontend engineer.

**Task:**
Create a production-ready React project for SynapseSignal dashboard.

**You must:**

- Initialize React app
- Setup:
  - folder structure (components, services, hooks, utils)

- Install:
  - Leaflet
  - Axios
  - Tailwind CSS

**Output:**

- Complete project structure
- Setup commands
- Base App component

**Constraints:**

- Clean architecture
- No unnecessary dependencies

---

## 2️⃣ MASTER PROMPT — MAP INTEGRATION (LEAFLET)

You are implementing a map system.

**Task:**
Integrate Leaflet map in React.

**You must:**

- Load OpenStreetMap tiles
- Center map on city
- Enable:
  - zoom
  - pan

**Output:**

- React component rendering map

**Constraints:**

- Must be responsive
- No blocking renders

---

## 3️⃣ MASTER PROMPT — INTERSECTION MARKER RENDERING

You are building map overlays.

**Task:**
Display intersections on map.

**Input:**

- intersection_id
- latitude
- longitude

**You must:**

- Render markers
- Add click handlers

**Output:**

```json
{
  "intersection_id": "...",
  "lat": ...,
  "lon": ...
}
```

**Constraints:**

- No duplicate markers
- Efficient rendering

---

## 4️⃣ MASTER PROMPT — SIGNAL STATE VISUALIZATION

You are visualizing traffic signals.

**Task:**
Show signal state per intersection.

**Input:**

- signal state (RED/GREEN/YELLOW)

**You must:**

- Color-code markers
- Update in real-time

**Output:**

- Dynamic marker updates

**Constraints:**

- No UI flickering
- Smooth updates

---

## 5️⃣ MASTER PROMPT — INTERSECTION DETAIL PANEL

You are designing detail view.

**Task:**
Display detailed traffic info on click.

**You must show:**

- lane densities
- sector scores
- congestion level

**Output:**

- Sidebar or popup UI

**Constraints:**

- Must update dynamically
- Clean layout

---

## 6️⃣ MASTER PROMPT — EMERGENCY VEHICLE TRACKING

You are building tracking UI.

**Task:**
Render moving emergency vehicle.

**Input:**

- latitude
- longitude
- type

**You must:**

- Display marker
- Update position in real-time

**Output:**

- Animated marker

**Constraints:**

- No lag
- Smooth movement

---

## 7️⃣ MASTER PROMPT — ROUTE VISUALIZATION

You are visualizing paths.

**Task:**
Draw route on map.

**Input:**

- path (list of coordinates)

**You must:**

- Render polyline
- Highlight route

**Output:**

- Visible path on map

**Constraints:**

- Accurate path rendering

---

## 8️⃣ MASTER PROMPT — GREEN CORRIDOR VISUALIZATION

You are visualizing corridor system.

**Task:**
Display active green corridor.

**Input:**

- corridor intersections

**You must:**

- Highlight route in green
- Animate progression

**Output:**

- Animated corridor

**Constraints:**

- Must sync with backend
- No visual mismatch

---

## 9️⃣ MASTER PROMPT — ROUTE COMPARISON UI

You are designing analysis panel.

**Task:**
Compare multiple routes.

**Input:**

- routes with:
  - ETA
  - congestion

**You must:**

- Display:
  - best route
  - alternatives
- Highlight optimal

**Output:**

- Route comparison panel

**Constraints:**

- Clear differentiation
- No clutter

---

## 🔟 MASTER PROMPT — REAL-TIME DATA INTEGRATION

You are integrating backend data.

**Task:**
Connect frontend to APIs.

**You must:**

- Fetch:
  - traffic data

    - signal data
- Use:
    - polling or WebSocket

**Output:**

- Data service layer

**Constraints:**

- Handle latency
- Retry on failure

---

## 1️⃣1️⃣ MASTER PROMPT — CONTROL PANEL

You are building user controls.

**Task:**
Allow manual system interaction.

**You must:**

- Add:
    - override button
    - emergency simulation
- Send commands to backend

**Output:**

- Interactive UI controls

**Constraints:**

- Must validate inputs
- Prevent unsafe commands

---

## 1️⃣2️⃣ MASTER PROMPT — DASHBOARD ANALYTICS

You are building analytics UI.

**Task:**
Display system insights.

**You must show:**

- congestion trends

- traffic stats
- system health

**Output:**

- Charts / panels

**Constraints:**

- Real-time updates
- Clean visuals

---

## 1️⃣3️⃣ MASTER PROMPT — UI OPTIMIZATION

You are optimizing frontend performance.

**Task:**
Improve responsiveness.

**You must:**

- Reduce re-renders
- Optimize map rendering
- Use memoization

**Output:**

- Optimized components

**Constraints:**

- No performance bottlenecks

---

## 1️⃣4️⃣ MASTER PROMPT — ERROR HANDLING UI

You are designing fallback UI.

**Task:**
Handle failures gracefully.

**You must:**

- Show:
  - loading states
  - error messages
- Handle API failures

**Output:**

- Robust UI states

**Constraints:**

- No crashes
- Clear user feedback

---

## 🔷 DEV 4 MASTER PROMPT OUTCOME

- Fully functional React frontend
- Map-based visualization system
- Real-time traffic & signal display
- Emergency tracking and route visualization
- Green corridor visualization
- Route comparison UI
- Backend integration layer
- Interactive control panel
- Analytics dashboard
- Optimized and robust UI

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✔ SynapseSignal Frontend becomes
👉 **a real-time city control interface**

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## 👤 DEV 4 — GUIDED LEARNING (Frontend + Map Intelligence | SynapseSignal)

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### 🔹 PHASE 1 — FRONTEND SETUP

#### 🧠 What You Are Learning

You are learning **how to structure a real-time frontend system**, not just a UI.

---

### ⚙️ What Is Happening

- React app is initialized
- Project structured into:
  - components
  - services
  - hooks

---

### 🧪 Guided Prompt

Run **React Setup Prompt**

---

### 🔍 What to Notice

- Clean structure → scalable system
- Messy structure → integration chaos later

---

### 🧱 Mental Model

**Frontend = system, not just visuals**

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## 🔹 PHASE 2 — MAP INTEGRATION

### 🧠 What You Are Learning

You are learning **how digital systems represent physical space**.

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### ⚙️ What Is Happening

- Leaflet loads real-world map
- Coordinates → visual positions

---

### 🧪 Guided Prompt

Run **Map Integration Prompt**

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### 🔍 What to Notice

- Lat/Lon directly maps to screen

- Map is base layer for all intelligence

---

### 🧱 Mental Model

**Map = canvas of your system**

---

## 🔹 PHASE 3 — INTERSECTION VISUALIZATION

### 🧠 What You Are Learning

You are learning **how backend nodes become visual entities**.

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### ⚙️ What Is Happening

- Each intersection becomes a marker
- Clickable and interactive

---

### 🧪 Guided Prompt

Run **Intersection Marker Prompt**

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### 🔍 What to Notice

- Incorrect coordinates → wrong placement
- Too many markers → performance issues

---

### 🧱 Mental Model

**Nodes in graph = markers on map**

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## 🔹 PHASE 4 — SIGNAL STATE VISUALIZATION

### 🧠 What You Are Learning

You are learning **how control decisions are visualized**.

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### ⚙️ What Is Happening

- Marker color reflects:
  - RED / GREEN / YELLOW

- Updates in real-time

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### 🧪 Guided Prompt

Run **Signal Visualization Prompt**

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### 🔍 What to Notice

- Real-time sync is critical
- Flickering = poor state handling

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### 🧱 Mental Model

**Color = system state**

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## 🔹 PHASE 5 — INTERSECTION DETAIL PANEL

### 🧠 What You Are Learning

You are learning **how to drill down into system state**.

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### ⚙️ What Is Happening

- Click → show:
  - lane metrics
  - sector scores
  - congestion

---

### 🧪 Guided Prompt

Run **Detail Panel Prompt**

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### 🔍 What to Notice

- Too much data = confusion
- Structure matters

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### 🧱 Mental Model

**UI must simplify complexity**

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## 🔹 PHASE 6 — EMERGENCY VEHICLE TRACKING

### 🧠 What You Are Learning

You are learning **real-time object visualization**.

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### ⚙️ What Is Happening

- Emergency vehicle moves on map
- Position updates continuously

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### 🧪 Guided Prompt

Run **Emergency Tracking Prompt**

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### 🔍 What to Notice

- Smooth motion vs jumping
- Latency affects realism

---

### 🧱 Mental Model

**Movement = continuous updates, not redraws**

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## 🔹 PHASE 7 — ROUTE VISUALIZATION

### 🧠 What You Are Learning

You are learning **how paths are rendered visually**.

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### ⚙️ What Is Happening

- Draw polyline for route
- Connect intersections

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### 🧪 Guided Prompt

Run **Route Visualization Prompt**

### 🔍 What to Notice

- Path accuracy matters
- Wrong order → broken route

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### 🧱 Mental Model

**Route = connected nodes**

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## 🔹 PHASE 8 — GREEN CORRIDOR VISUALIZATION

### 🧠 What You Are Learning

You are learning **how control strategy becomes visible**.

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### ⚙️ What Is Happening

- Highlight corridor path
- Animate signal progression

---

### 🧪 Guided Prompt

Run **Corridor Visualization Prompt**

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### 🔍 What to Notice

- Must match backend decisions exactly
- Visual mismatch → system distrust

---

### 🧱 Mental Model

**Visualization must reflect reality**

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## 🔹 PHASE 9 — ROUTE COMPARISON UI

### 🧠 What You Are Learning

You are learning **decision transparency**.

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### ⚙️ What Is Happening

- Show:
  - current route
  - alternatives
- Compare:
  - ETA
  - congestion

---

### 🧪 Guided Prompt

Run **Route Comparison Prompt**

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### 🔍 What to Notice

- Helps user trust system decisions
- Clear comparison improves usability

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### 🧱 Mental Model

**Explain decisions, don't hide them**

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## 🔹 PHASE 10 — REAL-TIME DATA INTEGRATION

### 🧠 What You Are Learning

You are learning **frontend-backend synchronization**.

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### ⚙️ What Is Happening

- Fetch data continuously
- Update UI state

---

### 🧪 Guided Prompt

Run **Data Integration Prompt**

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### 🔍 What to Notice

- Delays → outdated UI
- Too frequent calls → performance issues

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### 🧱 Mental Model

**Frontend reflects backend state continuously**

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## 🔹 PHASE 11 — CONTROL PANEL

### 🧠 What You Are Learning

You are learning **human-in-the-loop systems**.

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### ⚙️ What Is Happening

- User can:
  - override signals
  - simulate emergencies

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### 🧪 Guided Prompt

Run **Control Panel Prompt**

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### 🔍 What to Notice

- Must prevent unsafe inputs
- UI should guide user decisions

---

### 🧱 Mental Model

**User interaction must be controlled and safe**

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## 🔹 PHASE 12 — DASHBOARD ANALYTICS

### 🧠 What You Are Learning

You are learning **how to summarize complex systems**.

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### ⚙️ What Is Happening

- Display:
  - trends
  - congestion stats

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### 🧪 Guided Prompt

Run **Analytics Prompt**

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### 🔍 What to Notice

- Too much data overwhelms
- Trends > raw values

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### 🧱 Mental Model

**Insight > raw data**

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## 🔹 PHASE 13 — UI OPTIMIZATION

### 🧠 What You Are Learning

You are learning **performance engineering in frontend**.

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### ⚙️ What Is Happening

- Reduce re-renders
- Optimize map updates

---

### 🧪 Guided Prompt

Run **UI Optimization Prompt**

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### 🔍 What to Notice

- Maps are heavy components
- Optimization is necessary

---

### 🧱 Mental Model

**Performance = usability**

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## 🔹 PHASE 14 — ERROR HANDLING

### 🧠 What You Are Learning

You are learning **resilient UI design**.

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### ⚙️ What Is Happening

- Show:
  - loading
  - error states

---

### 🧪 Guided Prompt

Run **Error Handling Prompt**

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### 🔍 What to Notice

- UI must never crash
- Always provide feedback

---

### 🧱 Mental Model

**Failure must be handled gracefully**

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## 🔷 FINAL LEARNING OUTCOME

You now understand:

- How frontend connects to real-world geography
- How backend data becomes visual intelligence
- How maps represent system state
- How emergency tracking is visualized
- How routes and corridors are displayed
- How real-time systems update UI
- How to design interactive control panels

- How to optimize performance
- How to build resilient interfaces

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## 🧠 FINAL INSIGHT

> Dev 4 turns invisible system intelligence
> into **visible, understandable, and controllable reality**

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✔ Ready for **Dev 5 (Database Layer)**